

TABLE I
7-BIT ASSUMPTION
FOR ALL CHARACTERS

Character	Binary representation
A	011 0001
B	011 0010
C	011 0011
D	011 0100
E	011 0101
F	011 0110
G	011 0111
H	011 1000
I	011 1001
J	011 1010
K	011 1011
L	011 1100
M	011 1101
N	011 1110
O	011 1111
P	100 0000
Q	100 0001
R	100 0010
S	100 0011
T	100 0100
U	100 0101
V	100 0110
W	100 0111
X	100 1000
Y	100 1001
Z	100 1010

on at an extremely low time delay (a few milliseconds), so that the process cannot be detected by human eyes.

The concept of VLC is similar to infrared (IR). Any kind of light source with faster switching speed can be used as the transmitting device for VLC. However, some are better suited than others. For instance, incandescent lights quickly break down when switched on and off frequently. Thus, these are not recommended to be used as VLC transmitters. More promising alternatives are fluorescent lights and LEDs.

Fluorescent lights are usually used for illumination. But these can also be used as transmitters, because these can flicker well enough to transmit a meaningful amount of data.

LEDs are replacing fluorescent lamps, since LEDs are highly power-efficient, and have a longer lifespan. This is a major advantage for VLC communication. Since LEDs have a

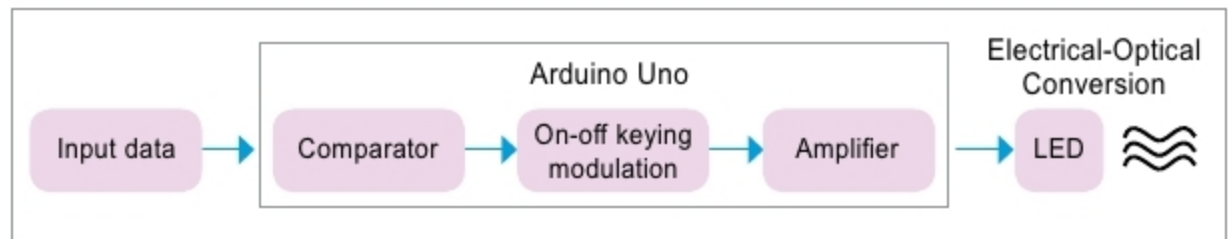


Fig. 4: Block diagram of VLC transmitter

higher flicker rate than fluorescent lights, rate of data transmission is increased greatly.

Development of LEDs has made solid-state lighting an emerging field. LEDs have surpassed incandescent light sources in terms of reliability, power requirements and luminous efficiency. Efficiency of LEDs is much higher than that of incandescent lamps. LEDs and lasers are used as transmission sources for VLC. White light-based LEDs are an attractive candidate for being used as VLC source.

There are different spectra in which white light is produced by LEDs. Tetra-chromatic (combination of cyan-red-green-blue), dichromatic (combination of blue-yellow) and trichromatic (combination of red-green-blue) are the different modes used for the generation of white light. The most common method for generating white light using LEDs is trichromatic. For more efficiency, monolithic trichromatic LEDs are used, since these have high data transfer speed as compared to other LED types.

Input data (character) to be sent from a personal computer is first converted into corresponding binary code, in which 1s in the binary code represent LED on. Similarly, 0s represent LED off. The character is represented in binary form by the MCU in Arduino. Because of the high on-off speed characteristic of LEDs, humans cannot perceive the twinkling phenomena, so both lighting and information transmitting can be realised simultaneously. Generated optical signals carrying original information are then delivered to the indoor wireless channel.

In the above example, seven bits are assigned to each character [A-Z] in Arduino. The program for transmission is compiled and uploaded to Arduino, which controls the light at

transmitter end and sends the compiled information. The 7-bit assumption for each character is given in Table I.

In binary representation, logic 1 represents LED on and logic 0 represents LED off. This modulating technique is known as OOK. Here, the LEDs are not completely turned off in off state, but the level of intensity is reduced.

The main advantage of using OOK is easy implementation. Other modulating techniques can also be used for data transmission. Each modulating technique has its own characteristics. Some other modulating techniques are PWM, M-PAM and M-PPM.

VLC receiver

A receiver module contains photodiode, Arduino and PC. Light from the transmitter is picked up by a single, or an array of, photosensitive filament on receiver side. In VLC receiver, light is detected using a photosensitive device and then converted to photo current.

Parameter specification of VLC is different from that of IR communication because of different wavelengths. When subjected to light energy, a photoconductive light sensor changes its physical property. A photoresistor (light-dependent resistor, LDR) is the most commonly used photosensitive device. It is a semiconductor device that uses light energy to control the flow of electrons and therefore the flow of current in these.

An LDR changes its electrical resistance depending on the intensity of light. It changes its electrical resistance from a high value of several thousand Ohms in the dark to only a few hundred Ohms when light is incident on it by creating electron-hole pairs in the material.

LDR is preferred because it has low rise and moderate fall times. Faster rise and fall times are essential